

B.A./B.Sc. (Fifth Semester)

Examination, 2024-25

MATHEMATICS

Paper - I (Theory)

[Abstract Algebra]

Time : 3 Hours]

[Maximum Marks : 75

Note : This question paper is divided into two sections A and B. Attempt any *five* questions each of 6 marks from the section A and any *three* questions each 15 marks from Section B.

SECTION-A

(Short Answer Type Questions)

Note : Attempt any *five* questions from this section. Each question carries equal marks. (5×6=30)

1. Show that the set $S = \{1, 5, 7, 11\}$ is a group with respect to the multiplication module 12.

2. Prove that every cyclic group is abelian. Show by an example that converse may not be true in general.
3. Let G be the group of real numbers under addition and let $\bar{G}$ be the group of positive real numbers under multiplication. Show that G and $\bar{G}$ are isomorphic under $\phi : G \rightarrow \bar{G}$, $\phi(x) = 2^x$.
4. Show that every subgroup of an abelian group is normal.
5. Let R be a commutative ring. Then prove that R is integral domain if and only if the cancellation laws hold in R .
6. Prove that the intersection of any two subrings of a ring R is a subring of R .
7. Let R be the ring of all real valued continuous functions on $[0, 1]$. Show that the set $S = \left\{ f \in R : f\left(\frac{1}{2}\right) = 0 \right\}$ is an ideal of R .
8. Define the following terms :
 - (a) Characteristics of a Ring.
 - (b) Center of the group.
 - (c) Units in a ring.

SECTION-B

(Long Answer Type Questions)

Note : Attempt any *three* questions from this sections. Each question carries equal marks. (3×15=45)

9. Show that the collection A_3 of even permutation forms a subgroup of the symmetric group S_3 with respect to the permutation multiplication. Check whether A_3 is a normal subgroup of S_3 .

10. Prove that, for all element a and b of the group G

$$(ab)^{-1} = b^{-1}a^{-1}.$$

11. State and prove Lagrange's theorem.

12. Show that the set

$$\mathbb{Z}[i] = \{a + ib : a, b \in \mathbb{Z}\}$$

forms a ring under ordinary addition and multiplication of complex numbers, where $i = \sqrt{-1}$. Is $\mathbb{Z}[i]$ an integral domain?

13. Prove that a finite integral domain is a field.
